

CC262 - Information Security

Credit Hours: 3(3+0) **Prerequisites:** None

Course Category: Core Computing Course

Description

This course provides a broad overview of the threats to the security of information systems, the responsibilities and basic tools for information security, and the levels of training and expertise needed in organizations to reach and maintain a state of acceptable security. It covers concepts and applications of system and data security. Areas of particular focus include secure network design, implementation and transition issues, and techniques for responding to security breaches.

Textbook(s)

- Computer and Information Security Handbook (4th Ed.), edited by John R. Vacca, Morgan Kaufmann (2024), ISBN: 978-0443132230.

- William Stallings, Lawrie Brown,. Computer Security: Principles and Practice 4th Edition, Pearson, (2017), ISBN: 978-0134794105

- Michael E. Whitman, Herbert J. Mattord.Principles of Information Security 6th Edition, Cengage Learning, (2017), ISBN: 978-1337102063

Course Learning Objectives

#	Course Learning Outcomes (CLOs)	BT/ PLO #
CLO 1	Explain key concepts of information security such as design principles, cryptography, risk management, and ethics	C2 1,10
CLO 2	Discuss legal, ethical, and professional issues in information security	A2 2,3,7 8,9
CLO 3	Apply various security and risk management tools for achieving information security and privacy	C3 4,5
CLO 4	Identify appropriate techniques to tackle and solve problems in the discipline of information security	C4 2,3,4

Week wise Lecture Schedule

W #	Lecture Topics
1	Information security Fundamentals: Define Security, Role & Significance, Applications, Beneficiaries, Characteristics of Information i.e. Confidentiality, Integrity, Availability, Authenticity, Utility etc, Terminologies of Information Security
2	Memory exploits, i.e. Buffer overflow (Stack Overflow, Heap Overflow, Integer Overflow) , Format String

W #	Lecture Topics
3	Threats, its Types & Defenses. Virus, Malware, Spyware, Trojan, Phishing, Ransomware, Breaches etc.
4	Attacks, its Types & Defenses. Masquerade, Modification, Repudiation, Eavesdropping, Spoofing, DoS etc.
5	Crimes, its Types & Defenses. Crime, Computer Crime, Cyber Crime. Types of Cyber Crimes: Financial Fraud, Counterfeiting, ID Theft, Cyber Terrorism, Cyber Extortion, Cyber Warfare, Crypto Jacking, Cyber Bullying, Cyber Talking, Cyber Espionage, Cyber Squatting, Digital Piracy, IP Infringement, Privacy Invasion, Doxing, BEC & Exit Scam, Spam & Salami, Hacking, Botnet & Predator etc. Cyber Crimes Prevention and Combating Approaches
6	System Security: Defining System Security and its Objectives. Violations via Vulnerabilities, Threats, and Attacks. System Breaches via Confidentiality, Integrity, Authentication, ToS, and DoS
7	Information security Policies, Standard and practices . Types of security policies (Access control lists, Access control Metrix, Configuration Rules)
8	Access Control, Its types and Elements of Access Control . Access control Models (Mandatory access control, Nondiscretionary access control, Discretionary access control)
9	Network Security
10	Cryptography, Definitions and Concepts,
11	Cipher Methods (Bit Stream, Block Stream). Substitution Cipher (Monoalphabetic substitution, Polyalphabetic substitution, Vigenère cipher)
12	Transposition Cipher (Caesar Matrix, Exclusive OR (XOR), Varnam Cipher, Book or Running Key Cipher)
13	Symmetric Key Cryptography. Private Key Distribution Approaches. Stream Ciphers vs. Block Ciphers. DES Algorithm vs. AES Algorithm
14	Asymmetric Key Cryptography. Public Key Distribution Approaches. RSA Algorithm vs. PGP Application. Comparison of Symmetric and Asymmetric Key Cryptography
15	Intrusion Detection and Prevention Systems
16	Signature based IDPS, Behavior Based IDPS, Network-based IDPS, Host-based IDPS